

Creating and using RDS.

1. What is RDS?

Amazon RDS

Amazon RDS (Relational Database Service) is a **managed service by AWS** that lets you run relational databases without handling infrastructure.

In simple terms:

You focus on data → AWS handles server, OS, maintenance.

2. Why RDS Exists (Core Idea)

Without RDS:

- Install DB manually
- Configure backups
- Handle crashes
- Patch OS

With RDS:

- AWS automates all of it

Creation Of RDS.

Create database Info

 **Free plan has access to limited features and resources**
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Upgrade plan 

Choose a database creation method

☒ **Full configuration**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type Info

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ MySQL


☐ PostgreSQL


☐ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Edition

Then creation of db name , username and password.

The screenshot shows the 'Create database' wizard in the AWS RDS console. The 'Credentials Settings' section is active, showing fields for 'Master username' (admin), 'Master password' (with a strength indicator), and 'Confirm master password'. Below this, the 'Additional credentials settings' section shows 'Database authentication options' with 'Password authentication' selected. The 'Instance type' section shows 'db.t3.micro' selected. The 'Storage' section shows 'General Purpose SSD (gp2)' selected and 'Allocated storage' set to 400 GiB. The 'Connectivity' section shows 'Don't connect to an EC2 compute resource' selected.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)
Password strength: **Very weak**
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)
Password strength: **Very weak**

Additional credentials settings

Database authentication options [Info](#)
☒ Password authentication
Authenticates using database passwords.
☐ Password and IAM database authentication

Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
400 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Created Database.

The screenshot shows the 'Databases' page in the AWS RDS console. A table lists the database instances. The table has columns for DB identifier, Status, Role, Engine, Upgrade rollout order, Region, and Size. One instance, 'database-1', is listed with a status of 'Available'.

Aurora and RDS > Databases

Databases (1)

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region	Size
database-1	Available	Instance	MySQL Co...	SECOND	us-east-1b	db.t3.micro

Read Replica.

Then we have to Create an Read Replica.

It Will allow you to duplicate your data when it acts as secondary database when primary db fails it acts as an primary database.

Settings

Replica source
Source DB instance identifier
database-1
Role: Instance

DB instance identifier
This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).
DATABASE2

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | Info
▼ Hide filters

☐ Include previous generation classes

☐ Standard classes (includes m classes)
☐ Memory optimized classes (includes r and x classes)
☒ Burstable classes (includes t classes)

Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

This is the Created Read Replica.

Aurora and RDS <

Databases (2) Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
database-1	Available	Primary	MySQL Co...	SECOND	us-east-1b	db.t3.micro
database2	Available	Replica	MySQL Co...	SECOND	us-east-1b	db.t3.micro

Then we have to connect to Database through the instance.

Command -- `mysql -h your-endpoint.rds.amazonaws.com -u admin -p`.

```

Setting up libhttp-message-perl (6.45-1ubuntu1) ...
Setting up mysql-server (8.0.45-0ubuntu0.24.04.1) ...
Setting up libcgi-pm-perl (4.63-1) ...
Setting up libhtml-template-perl (2.97-2) ...
Setting up libcgi-fast-perl (1:12.17-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-21-232:/home/ubuntu# mysql -h database-1.cq56uimsqe8t.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 30
Server version: 8.4.7 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>

```

i-00f6cc7902f8f0748 (ubuntu)

PublicIPs: 13.218.242.161 PrivateIPs: 172.31.21.232

Then I have created one database and Created one table.

```

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.01 sec)

mysql> CREATE DATABASE HARIDB;
Query OK, 1 row affected (0.04 sec)

mysql> USE HARIDB;
Database changed

```

```

mysql> USE HARIDB;
Database changed
mysql> CREATE TABLE Persons (
  ->   PersonID int,
  ->   LastName varchar(255),
  ->   FirstName varchar(255),
  ->   Address varchar(255),
  ->   City varchar(255)
  -> );
Query OK, 0 rows affected (0.05 sec)

```

```

mysql> DESC Persons;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| PersonID | int | YES | | NULL | |
| LastName | varchar(255) | YES | | NULL | |
| FirstName | varchar(255) | YES | | NULL | |
| Address | varchar(255) | YES | | NULL | |
| City | varchar(255) | YES | | NULL | |
+-----+

```

i-00f6cc7902f8f0748 (ubuntu)

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